

Distal and Local Group Learning: Performance Trade-offs and Tensions

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There is increasing recognition that group members learn not only within the group (i.e., local learning), but also externally (i.e., distal learning), and these two group learning processes may facilitate group performance in different ways. Yet, despite this recognition, there is much that is not understood about whether they complement or inhibit each other in affecting group performance, and whether group social and task conditions that foster one type of learning do so at the expense of the other. The findings from this field study of teams from four firms show that (1) local learning and distal learning are positively related to group efficiency and group innovativeness, respectively; (2) distal learning negatively interacts with local learning to impede group efficiency; and (3) high levels of group cohesion promote distal learning but diminish local learning. Overall, these findings suggest that there are not only performance trade-offs to engaging in either only local or distal learning, but also performance disadvantages to engaging in both types of group learning because distal learning impedes local learning from achieving a high level of group efficiency. In addition, there is preliminary evidence to suggest that tensions can arise from simultaneously managing both types of group learning because a high level of group cohesion increases distal learning but decreases local learning.

Key words: group learning; group innovativeness; group efficiency

Defined as the knowledge acquisition, sharing, and combination activities of group members (Argote et al. 2001), group learning has been posited to advance group development (Argote et al. 2001), enhance group creativity (Hargadon 1999), coordinate members' thoughts and actions (Senge 1990), and promote group performance (Edmondson 1999). Research in this area has particularly underscored the importance of distinguishing between group learning processes that refine and improve current capabilities and group learning processes that discover and develop new capabilities because they are likely to impact group performance in different ways (e.g., Bunderson and Sutcliffe 2003, Edmondson 2002).

Extending this line of research, this study focuses on group learning that takes place with individuals in the immediate group (referred to as *local* learning in this study) and group learning that occurs with individuals external to the group (referred to as *distal* learning in this study). This distinction is not new as studies have demonstrated that group members turn to both internal and external interpersonal sources for knowledge exchange (e.g., Allen 1970, Ancona and Caldwell 1992a). Research suggests that while learning within the group builds shared understanding and aligns collective actions (Fiol 1994, Senge 1990), learning with external individuals acquires and develops novel knowledge (Hansen 1999, Sutton and Hargadon 1996).

The different focus of local and distal learning invites questions about whether they complement or impede

the other in accomplishing group tasks. There is general recognition that different types of group learning may promote group performance in different ways (e.g., Edmondson 2002), but there has been little research on the interactive effects of local and distal learning on group efficiency and group innovativeness. As Argote et al. (2001) noted, there is often the need to simultaneously manage different group learning processes (e.g., transferring knowledge from other groups and combining knowledge of group members) and, so, it is important to investigate whether there are any performance trade-offs from emphasizing one type of group learning over the other and/or engaging in both types of group learning at the same time. For example, local and distal learning may mutually hinder each other such that group performance is diminished when both learning processes occur at the same time. In addition, given that local learning involves internal knowledge sharing and distal learning involves external knowledge sharing, tensions from fostering both types of learning may arise because group conditions that promote one type of group learning may do so at the expense of the other. Hence, there is a need for greater investigation into whether local and distal learning are differentially related to the social and task conditions of the group.

In view of these research gaps, the two primary goals of this study are to examine (1) how local and distal learning are independently and jointly related to group innovativeness and efficiency and (2) whether group

social and task conditions have differing relationships with local and distal learning. The terms group and team are used interchangeably in this paper to refer to a group of individuals in an organization with clearly defined membership, who are responsible for a shared product or service (Hackman 1987).

Theoretical Background and Hypotheses

Following Edmondson (1999) and Argote et al. (2001), this study adopts a process approach to examining group learning, that is, learning activities such as dialogue (Senge 1990), group reflection (Edmondson 1999), and brainstorming (Sutton and Hargadon 1996). Local learning is defined as the interpersonal knowledge acquisition, sharing, and combination activities with *members in the same group*, and distal learning is defined as the interpersonal knowledge acquisition, sharing, and combination activities with *individuals external to the group*. Similar to Cyert and March's (1963) distinction of local and distant search where the type of search depends on how close or far the search for solutions is from the vicinity of the problem, local and distal learning characterize whether group members engage in learning activities with local individuals (i.e., within the group membership boundary) or distant individuals (i.e., outside the group membership boundary).

Independent and Interaction Effects of Distal and Local Learning

A key theoretical premise of this paper is that local and distal learning promote different dimensions of group performance because they are likely to access and create different types of knowledge in the group. A strong theme in the organizational learning literature is that learning that is focused on exploiting preexisting knowledge bases enhances efficiency, and learning that is focused on exploring new knowledge bases promotes innovativeness (March 1991). Similarly, as individuals in the same immediate social circle tend to possess similar information and individuals outside of one's social circle tend to possess unique information (e.g., Granovetter 1983), we can expect that group members are accessing and creating more similar knowledge in local learning and more diverse knowledge in distal learning.

Accordingly, I hypothesize that distal learning promotes group innovativeness and local learning promotes group efficiency. Distal learning is likely to enhance group innovativeness because it not only acquires novel knowledge from external members (e.g., Ancona and Caldwell 1992a, b), but can also stimulate group members to consider how different knowledge can be combined to create an innovative solution (Sutton and Hargadon 1996). As research on minority influence has shown, different views of a minority stimulate "divergent thinking" in a group, where "[a]ttention is widened

to include other aspects of the stimuli . . . and, as a result, 'novel' solutions or decisions are often made" (Nemeth and Kwan 1987, p. 790). For example, a group is able to detect more correct novel solutions (Nemeth and Wachtler 1983) and create more original associations to words (Nemeth and Kwan 1985). Similarly, the diverse knowledge acquired as a result of distal learning is likely to trigger divergent thinking and, thus, facilitate the creation of novel knowledge (e.g., new ideas and solutions).

HYPOTHESIS 1A. *Distal learning is positively related to group innovativeness.*

Unlike the differentiation-enhancing nature of distal learning, local learning is likely to be coherence building—creating and maintaining shared group knowledge (i.e., knowledge about tasks, group processes, group members) that promotes greater group efficiency. For example, greater learning within the team can create greater shared understanding that aligns collective action (Fiol 1994, Senge 1990), build a "community memory" of task knowledge that facilitates speedy diagnosis and resolution of problems (Orr 1990), and develop a "trans-active memory system" or shared knowledge about each other's expertise that enhances more efficient coordination of members' expertise (Liang et al. 1995). By developing shared group knowledge that improves coordination and reduces misalignments, local learning is likely to enhance a group's ability to execute its tasks on schedule and within budget. Although the combination of different members' knowledge in local learning creates the potential for novel understanding to be discovered, the similar task and team mental models of individuals within the same group suggest that any "new" understanding that is produced is likely to be in the neighborhood of the current solution. Consequently, local learning is likely to deepen and develop greater shared understanding about current tasks and how tasks are currently executed in the group.

HYPOTHESIS 1B. *Local learning is positively related to group efficiency.*

In the case of group innovativeness, I hypothesize that distal and local learning complement one another to enhance group innovativeness (i.e., a positive interaction effect). As a group engages in a greater level of distal learning, more divergent thinking and combinatorial possibilities are likely, thereby leading to higher group innovativeness (e.g., Hargadon 1999). However, as Sutton and Hargadon's (1996) case study showed, group members also have to actively engage in mutual learning (specifically, brainstorming) to consider how external knowledge that is acquired can be combined to create an innovative solution. Group members are likely to need local learning—sharing their ideas and knowledge—so as to apply and adapt the external ideas to the group context to produce a novel solution. As studies of innovation

adoption have suggested, recipients of new ideas and practices often have to adapt the ideas and practices to their context to be effective (e.g., Leonard-Barton 1988, Rice and Rogers 1980).

HYPOTHESIS 2A. *The higher the local learning, the more likely distal learning is positively associated with group innovativeness.*

In the case of group efficiency, I posit that distal learning impedes local learning from achieving group efficiency (i.e., a negative interaction effect). As previously highlighted, distal learning is likely to trigger divergent thinking, which in turn is likely to increase cognitive variation in members' beliefs about their tasks and how things are done. Therefore, higher levels of distal learning probably counteract the positive impact of local learning on group efficiency because the greater cognitive differences that result hamper coordination and lead to greater misalignments in actions. As group heterogeneity and conflict research illustrates, divergent perspectives in a group reduce social integration (e.g., O'Reilly III et al. 1989) and increase conflict (e.g., Jehn 1995). Because of the introduction of greater cognitive variation, higher levels of distal learning are likely to not only increase the time required for local learning to create cognitive and behavioral convergence, but also reduce the success rate of building convergence as divergent perspectives and actions grow in the group. Thus, by impeding local learning from building the shared understanding necessary for efficient execution of group tasks, distal learning is likely to negatively interact with local learning to impact group efficiency.

HYPOTHESIS 2B. *The higher the distal learning, the less likely local learning is positively associated with group efficiency.*

The Differential Impacts of Social and Task Environments on Distal and Local Learning

There is growing evidence that social relationships facilitate interpersonal and intergroup knowledge sharing. Socially tied individuals "have greater motivation to be of assistance and are typically more easily available" (Granovetter 1983, p. 209), experience greater cognitive ease in transferring knowledge (Hansen 1999), and develop greater trust to discuss ideas with others (Albrecht and Hall 1991). At the group level, for instance, greater perceived psychological safety and trust have been shown to promote greater learning in the team (Edmondson 1999). Therefore, stronger group cohesion, which implies greater solidarity, trust, and mutual liking, is likely to foster greater local learning in the group by increasing the motivation, trust, and cognitive familiarity for productive inquiry.

HYPOTHESIS 3A. *Group cohesion is positively related to local learning.*

With respect to distal learning, group cohesion is a double-edged sword. On one hand, group cohesion is strongly associated with group identity (Jackson and Smith 1999), where members become more likely to cooperate with each other and commit to the group (see Turner et al. 1987). As such, they are likely to be motivated to enhance group performance (e.g., Harkins and Szymanski 1989) and, therefore, pursue distal learning to improve the group's practices and solutions. On the other hand, high levels of group cohesion promote insularity and resistance to external stimuli (Janis 1972). As social identity theory suggests, excessive levels of group identity lead to strong bias in favor of in-groups against salient out-groups (Tajfel and Turner 1979). Accordingly, I posit an inverted U-shaped relationship between group cohesion and distal learning. While some degree of group cohesion is necessary to motivate distal learning, excessive levels become ineffective.

HYPOTHESIS 3B. *There is an inverted U-shaped relationship between group cohesion and distal learning in the group.*

Just as social integration differentially affects local and distal learning, I posit that local and distal learning are related in different ways to group task. As a key dimension of work unit technology (Perrow 1967), task routineness is measured by the frequency of unexpected and novel events that occur in the accomplishment of a task. When task routineness decreases, members experience greater changes and exceptions to their task and, hence, are likely to become less familiar with their task and have less slack—in terms of time resources—to "fight fires." Some variation in tasks is, however, beneficial for group learning as related task variation helps group members to develop a deeper understanding of the concepts underlying their tasks (Schilling et al. 2003).

The extent to which a group engages in local or distal learning is likely to vary with the degree of task routineness in the group. As greater boundary spanning helps the team to obtain external knowledge to deal with novel issues (e.g., Ancona and Caldwell 1992a), members have to search outside of the group for information and knowledge to help resolve any exceptions that arose from novel and unexpected task changes (see Leifer and Delbecq 1978). Although it is possible that groups might seal off new information in response to perceived threats from unexpected and unfamiliar task demands (e.g., Janis 1972, Staw et al. 1981), the perceived need to acquire novel knowledge is likely to drive groups to engage in more distal learning as task routineness decreases.

HYPOTHESIS 4A. *Distal learning is negatively related to task routineness.*

Conversely, when task exceptions rarely occur, the relative familiarity with the task reduces the need for members to engage in distal learning. Instead, the greater task

certainty calls for more “first-order learning,” that is, exploiting familiar skills for addressing familiar problems (e.g., Sitkin et al. 1995). To the extent that task routineness accentuates the need to develop effective routines to maximize efficiency (e.g., Perrow 1970), increasing task routineness is likely to motivate members to engage in more frequent local learning. In addition, as task routineness increases, greater slack resources (especially time) and members’ greater familiarity with group tasks are likely to provide the time and shared knowledge for local learning. Greater shared understanding about the task and each other’s roles creates the common ground for cooperation, coordination, and communication (Mathieu et al. 2000). While it is conceivable that the absence of interruptions in task routineness leads to mindlessness and, therefore, inhibits local learning, this negative association between task routineness and local learning is only likely to occur when there is high group tenure (e.g., Katz 1982) or highly “habitual” group routines (e.g., Gersick and Hackman 1990). Therefore, I hypothesize that task routineness is positively associated with local learning.

HYPOTHESIS 4B. *Local learning is positively related to task routineness.*

Research Methodology

A field study of teams from four different firms was conducted to empirically test the hypotheses. In total, 78 teams participated in this study, including 16 teams from a financial services firm, 28 teams from a hospital, 20 teams from a diversified industrial firm, and 14 teams from a high-technology firm. A diverse array of teams was sampled, ranging from functional to cross-functional teams and from new product development to manufacturing teams. Team managers and team members of participating teams completed a Web-based team manager questionnaire and team member questionnaire (created using the Survey SaidTM survey software), respectively.

Using a team member response rate criterion of 50%, 3 teams from the hospital, and 1 team each from the diversified industrial firm and high-technology firm were dropped. The final sample size comprised 73 teams, where the average response rate of each team was 83.5%. The team size ranged from 3 to 25 members (mean = 8.4). The average team tenure ranged from 3 months to 16 years (mean = 3.0 years).

Measures

Local Learning. Edmondson’s (1999) measure of team learning was adapted to create the local learning measure. As local learning measures knowledge acquisition, sharing, and combination activities within the group, group members’ knowledge exchange interactions with one another, such as seeking each other’s

perspectives, adjusting to each other’s ideas, reflecting, and exploring one another’s ideas, were included. An 11-item measure, scaled on a 5-point Likert scale (1 = not at all; 3 = to some extent; 5 = to a very great extent), was developed.

To ensure that this construct was distinct from traditional interaction frequency and group cooperation constructs, it was pilot tested with 14 teams, 12 of which were executive MBA student teams and 2 of which were the work teams of 2 executive MBA students. Members of these teams were asked to fill out the local learning scale, a frequency of communication scale, and a positive group interaction scale, which was adapted from Watson et al. (1991). Results of the exploratory factor analysis revealed that each item loaded on its respective factor, thus supporting the discriminant validity of the local learning measure.¹

Using the responses of team members in this study, both exploratory and confirmatory factor analyses were conducted on all multi-item measures used. The results are reported in the next subsection. Based on these results, three items of the local learning measure were dropped, leaving an eight-item measure. This measure and the other multi-item measures are reported in the appendix.

Distal Learning. As distal learning measures the knowledge acquisition, sharing, and combination activities with individuals external to the group, Ancona and Caldwell’s (1992b) scout activities scale in the boundary-spanning measure (α not reported in the study) and Sutcliffe’s (1994) organizational scanning measure ($\alpha = 0.88$) were adapted to create the distal learning measure. Examples of the items included are “our team seeks ideas/expertise from people external to the team,” and “our team reviews the team’s work with people external to the team.” The adapted five-item measure was scaled on a 5-point Likert scale, where 1 = not at all; 3 = to some extent; and 5 = to a very great extent.

Group Cohesion. The social integration scale ($\alpha = 0.85$) of O’Reilly III et al. (1989) was used to measure group cohesion. This four-item scale measures the degree to which members have strong social relationships with one another. Responses ranged from 1 to 5, where 1 = not at all; 3 = to some extent; and 5 = to a very great extent.

Task Routineness. An adaption of Perrow’s (1970) routineness index was used. The adapted measure consisted of five items, and responses were scaled on a 5-point Likert scale (1 = not at all; 3 = to some extent; 5 = to a very great extent).

Group Efficiency and Innovativeness. Following the Ancona and Caldwell (1992b) and Pelled et al. (1999) measures of efficiency of team operations and innovativeness of the team, team managers were asked to evaluate group performance in terms of group efficiency and

group innovativeness along a 5-point Likert scale, where 1 = poor; 3 = average; and 5 = superior.

Control Variables. Because the teams were from different firms, they were likely to be operating in different structural and cultural organizational environments. Differences in these environments likely led to different norms and opportunities for group learning, as well as differing performance expectations and demands. Hence, dummy variables were created for each firm to be included as control variables.

At the group level, group size, group tenure, and group heterogeneity were included as control variables. The group literature has highlighted that size significantly affects group dynamics and group performance (e.g., Brewer and Kramer 1986). For example, as group size expands, there is a greater likelihood of social loafing and free-riding, thereby decreasing the extent of local learning in a group. Similarly, group tenure is likely to affect the extent to which a group attends to outside information sources (Katz 1982) and, therefore, the degree of distal learning in the group. Group heterogeneity influences the degree of group conflict by affecting the degree of cognitive (dis)similarity among members and, therefore, the performance of a group (Pelled et al. 1999). In addition, to the extent that heterogeneous members have greater diverse external contacts (Ancona and Caldwell 1992b), greater group heterogeneity is likely to lead to greater distal learning. Group size was measured by the list of members provided by each team manager, group tenure by the average length of time each member has been a member of his or her team, and group heterogeneity by Blau's (1977) functional heterogeneity index to compute the distribution of educational backgrounds (in terms of areas such as accounting, engineering, finance, computer sciences, etc.) across members in the team.

Factor Analysis

Because modified scales were used in this study, both exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) were conducted on the multi-item measures—local learning, distal learning, group cohesion, and task routineness. Using the split-half sample cross-validation approach, the sample was randomly split into halves, where each subset was used for EFA and CFA, respectively. The subsets were obtained by first categorizing the total member sample ($n = 496$) according to members' respective teams and then randomly assigning members in each team to one of the two subsets. This helped to ensure some equivalence between the two subsets. The two subsets ($n = 248$) were separately used to conduct EFA and CFA.

A principal axis factoring and direct oblimin rotation EFA was conducted. Results of the EFA revealed four factors with eigenvalues greater than one. All items

loaded as expected, except for three local learning items and one task routineness item. After these items were dropped, including a scanning item that had a factor loading of only 0.25, the factor loadings of all remaining items were greater than 0.40. A CFA was conducted to verify the factor structure that emerged from the EFA. Findings from the CFA showed that the measurement model was a reasonable fit (standardized RMSR = 0.07; CFI = 0.94) and was a significant improvement over the independence model ($\Delta\chi^2 = 4,124.01$, $p < 0.001$). Following Anderson and Gerbing (1988), tests of convergent validity and discriminant validity were conducted. Convergent validity was high, as each indicator's estimated pattern coefficient on its hypothesized construct factor was significant and greater than twice its standard error. Discriminant validity was also high, as in each of the pairwise tests, the chi-square difference between the unconstrained and the constrained model was significant at $p < 0.001$.

Aggregation of Individual to Group Measures

The interrater agreement index, $r_{wg(j)}$ (James et al. 1984), and the intraclass correlation coefficient, ICC (Bliese and Halverson 1998) were used to test whether individual members' scores for the group-level variables could be aggregated to the group. The mean $r_{wg(j)}$ was 0.93 for local learning, 0.84 for distal learning, 0.88 for group cohesion, and 0.84 for task routineness, demonstrating that there was high within-group agreement. The ICC was 0.22 for local learning, 0.12 for distal learning, 0.23 for group cohesion, and 0.20 for task routineness. The one-way ANOVA analyses from which the ICC values were derived were significant ($p < 0.001$), suggesting that the ICC values were significantly different from zero (Kenny and LaVoie 1985). Overall, these results provided strong evidence of adequate within-group agreement and between-group differentiation, thereby supporting the aggregation of the scores to the group level.

Statistical Analytic Tests

Because there are multiple dependent variables, a multivariate regression analysis was conducted to perform an omnibus test of the relationships between the independent and dependent variables. Independent univariate regressions would inflate Type I error rates when there is more than one dependent variable; as such, there is a risk of declaring one or more univariate regressions significant when no relationship actually exists. Interpretations of the significance of the univariate regressions can proceed only if the omnibus test is significant. Following the multivariate regression analysis, hierarchical regression analyses were conducted to examine the additional variance explained by different sets of independent variables. Sets of variables were entered sequentially, beginning with the control variables in Step 1, independent

variables in Step 2, and interaction terms in Step 3. Following Aiken and West (1991), interaction terms and their component variables were mean-centered prior to the regression analyses to minimize multicollinearity.

Results

Table 1 reports the means, standard deviations, correlations, and Cronbach alphas of the control, independent, and dependent variables. Reliabilities for the variables were high, ranging from 0.75 to 0.86. As shown, two of the three firm dummy variables were significantly related to group size, group tenure, and group heterogeneity, suggesting that different firms design and structure their groups in significantly different ways. Of the group demographic control variables, group size was negatively related to local learning and group cohesion, group tenure was positively related to task routineness, and group heterogeneity was negatively related to task routineness. As expected, larger groups tend to be less cohesive and less likely to engage in local learning, and groups that perform routine tasks tend to be older and more homogeneous.

Hypotheses 1a and 1b posit that distal learning and local learning are positively associated with group innovativeness and group efficiency, respectively. Hypotheses 2a and 2b posit interaction effects of distal and local learning on group innovativeness and group efficiency, respectively. A multivariate regression analysis was first conducted, where group innovativeness and group efficiency were regressed on the control variables, distal learning, and local learning. The omnibus null hypothesis that local learning is not associated with group performance was rejected at the significance level of 0.10 (Wilks' Lambda = 0.92, $F = 2.52$, $p = 0.09$). Similarly, the omnibus null hypothesis that distal learning is not associated with group performance was rejected at the significance level of 0.10 (Wilks' Lambda = 0.92, $F = 2.73$, $p = 0.07$).

Hierarchical regression analyses were conducted to examine Hypotheses 1a, 1b, 2a, and 2b (see Table 2). As

a group, the control variables accounted for a trivial and insignificant percentage of the variance in group innovativeness ($R^2 = 3.6\%$, $p > 0.10$) and group efficiency ($R^2 = 5.7\%$, $p > 0.10$). When distal and local learning were added to the regression models, they significantly increased the variance explained in group innovativeness ($\Delta R^2 = 10.9\%$, $p < 0.05$) and group efficiency ($\Delta R^2 = 9.9\%$, $p < 0.05$).² Consistent with Hypothesis 1a, distal learning was significantly and positively associated with group innovativeness ($\beta = 0.33$, $p < 0.05$). Hence, Hypothesis 1a was supported. Local learning was also significantly and positively related to group efficiency ($\beta = 0.35$, $p < 0.05$), therefore providing support for Hypothesis 1b. It is worth noting that distal learning was not significantly associated with group efficiency and local learning was not significantly associated with group innovativeness, suggesting that distal and local learning exhibit distinct performance functions.

Hypothesis 2a proposes that distal and local learning complement each other in increasing group innovativeness. Hypothesis 2b proposes that distal and local learning impede each other in increasing group efficiency. As shown in Table 2, factoring in the interaction term did not significantly increase the variance explained in group innovativeness ($\Delta R^2 = 0.6\%$, $p > 0.10$). Hence, Hypothesis 2a was not supported. However, the interaction term was negatively and significantly related to group efficiency ($\beta = -0.30$, $p < 0.05$). This interaction term significantly increase the variance explained in group efficiency by 7%.

Following Aiken and West (1991), I probed the interaction effect by regressing group efficiency on local learning at low (one standard deviation below the mean), medium (at mean), and high (one standard deviation above the mean) values of distal learning. Consistent with Hypothesis 2b, local learning was significantly associated with group efficiency when distal learning was at low ($\beta = 0.57$, $p < 0.01$) or medium ($\beta = 0.34$, $p < 0.01$) levels, and became insignificant when distal learning was at a high level ($\beta = 0.11$, $p > 0.10$). This

Table 1 Descriptive Statistics of Variables

	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12
1. Firm dummy 1	na	na	na											
2. Firm dummy 2	na	na	-0.38***	na										
3. Firm dummy 3	na	na	-0.31**	-0.43***	na									
4. Group size	8.42	4.65	-0.29*	0.11	0.23 ⁺	na								
5. Group tenure	2.98	3.44	-0.27*	0.13	0.28*	0.32**	na							
6. Group heterogeneity	0.44	0.25	0.32**	-0.14	-0.34**	-0.01	-0.04	na						
7. Local learning	3.45	0.41	0.12	0.34**	-0.15	-0.20 ⁺	0.03	-0.06	(0.86)					
8. Distal learning	3.03	0.45	-0.01	-0.00	-0.14	-0.12	-0.08	0.11	0.46***	(0.84)				
9. Group cohesion	3.76	0.46	0.13	0.08	-0.14	-0.34**	-0.19	-0.06	0.52***	0.32**	(0.86)			
10. Task routineness	3.02	0.46	-0.26*	0.42***	0.10	0.13	0.37**	-0.43***	0.26*	-0.14	0.08	(0.75)		
11. Group efficiency	3.73	0.81	0.18	-0.06	-0.01	-0.03	0.03	0.14	0.32**	0.18	0.22 ⁺	-0.06	na	
12. Group innovativeness	3.62	0.87	0.00	-0.03	0.14	-0.04	0.00	-0.08	0.19	0.29*	0.13	0.13	0.40**	na

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ⁺ $p < 0.10$.

Table 2 Hierarchical Regression Analyses of Group Innovativeness and Group Efficiency on Distal and Local Learning^a

Variables	Group innovativeness			Group efficiency		
	Control β (s.e.)	H1a β (s.e.)	H2a β (s.e.)	Control β (s.e.)	H1b β (s.e.)	H2b β (s.e.)
Firm dummy 1	0.139 (0.352)	0.203 (0.375)	0.229 (0.386)	0.214 (0.325)	0.097 (0.348)	0.190 (0.343)
Firm dummy 2	0.125 (0.328)	0.166 (0.369)	0.164 (0.371)	0.096 (0.303)	−0.094 (0.343)	−0.101 (0.330)
Firm dummy 3	0.246 (0.371)	0.310 (0.375)	0.300 (0.378)	0.132 (0.343)	0.056 (0.348)	0.024 (0.336)
Group size	0.055 (0.024)	0.106 (0.024)	0.102 (0.024)	−0.045 (0.022)	0.037 (0.022)	0.026 (0.021)
Group tenure	−0.073 (0.036)	−0.065 (0.034)	−0.050 (0.035)	0.057 (0.033)	0.051 (0.033)	0.105 (0.031)
Group heterogeneity	−0.037 (0.482)	−0.065 (0.463)	−0.087 (0.479)	0.139 (0.445)	0.140 (0.430)	0.065 (0.426)
Local learning		0.028 (0.354)	0.033 (0.356)		0.352* (0.329)	0.371* (0.317)
Distal learning		0.326* (0.281)	0.348* (0.290)		0.018 (0.261)	0.097 (0.258)
Distal learning $\times$ local learning			−0.087 (0.573)			−0.303* (0.510)
R^2	0.036	0.146	0.152	0.057	0.156	0.229
ΔR^2		0.109*	0.006		0.099*	0.073*
F	0.401	1.321	1.212	0.641	1.430	2.011+
df	6, 64	8, 62	9, 61	6, 64	8, 62	9, 6

* $p < 0.05$; + $p < 0.10$.

^a $N = 71$. The sample available for these regression analyses was reduced from 73 to 71 as 2 team managers did not fill in the team manager questionnaire.

finding implies that as distal learning increases, local learning becomes less positively related to group efficiency, thereby supporting Hypothesis 2b.

Hypotheses 3a, 3b, 4a, and 4b examine the relationships of group cohesion and task routineness with distal and local learning. Results of the multivariate regression analysis showed that the omnibus null hypothesis that group cohesion is not associated with group learning processes was rejected at a significance level of 0.01 (Wilks' Lambda = 0.80, $F = 7.97$, $p = 0.001$). However, the omnibus null hypothesis that task routineness is not associated with group learning processes was not rejected (Wilks' Lambda = 0.96, $F = 1.21$, $p = 0.306$), suggesting that task routineness is not associated with distal and local learning.

Results of the hierarchical regression analysis are shown in Table 3. Although firm dummy variables had no effect on distal learning, they were significantly associated with local learning. Specifically, the Scheffé test of post-hoc ANOVA revealed that teams from the hospital engaged in significantly higher levels of local learning than teams from the industrial diversified and the high-technology firm, and teams from the financial service firm engaged in significantly higher levels of local learning than teams from the high-technology firm. Group demographic characteristics had no effect on distal learning, but group size was negatively associated with local learning.

As expected, group cohesion was positively and significantly related to local learning ($\beta = 0.45$, $p < 0.001$),

Table 3 Hierarchical Regression Analyses of Distal and Local Learning on Group Cohesion and Task Routineness^a

Variables	Local learning			Distal learning		
	Control β (s.e.)	H3a & H4a β (s.e.)	β (s.e.)	Control β (s.e.)	H4b β (s.e.)	H3b β (s.e.)
Firm dummy 1	0.332* (0.144)	0.295* (0.128)	0.353** (0.128)	−0.228 (0.178)	−0.226 (0.172)	−0.204 (0.177)
Firm dummy 2	0.560** (0.134)	0.485** (0.127)	0.497** (0.124)	−0.171 (0.166)	−0.136 (0.171)	−0.131 (0.172)
Firm dummy 3	0.213 (0.150)	0.218 (0.132)	0.186 (0.130)	−0.226 (0.185)	−0.195 (0.178)	−0.208 (0.180)
Group size	−0.241* (0.009)	−0.121 (0.008)	−0.140 (0.008)	−0.149 (0.012)	−0.058 (0.011)	−0.066 (0.012)
Group tenure	0.047 (0.014)	0.076 (0.013)	0.119 (0.013)	−0.014 (0.017)	0.070 (0.018)	0.086 (0.018)
Group heterogeneity	0.001 (0.198)	0.056 (0.188)	−0.004 (0.190)	0.088 (0.245)	0.055 (0.254)	0.031 (0.264)
Group cohesion		0.448*** (0.088)	2.955* (1.088)		0.341* (0.119)	1.326 (1.510)
Task routineness		0.072 (0.109)	0.056 (0.107)		−0.153 (0.148)	−0.159 (0.149)
Group cohesion (squared)			−2.528* (0.147)			−0.993 (0.204)
R^2	0.269	0.452	0.486	0.062	0.167	0.172
ΔR^2		0.184***	0.034*		0.106*	0.005
F	4.043**	6.608***	6.628***	0.721	1.605	1.457
df	6, 66	8, 64	9, 63	6, 66	8, 64	9, 63

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

^a $N = 73$.

thereby supporting Hypothesis 3a. To test Hypothesis 3b, which posits a curvilinear relationship between group cohesion and distal learning, a squared group cohesion term was entered into the regression model. Contrary to expectations, the squared group cohesion term was not significantly related to distal learning ($\beta = -0.99$, $p > 0.10$); hence, Hypothesis 3b was not supported. However, a significant linear and positive relationship was detected between group cohesion and distal learning ($\beta = 0.34$, $p < 0.01$). Surprisingly, the squared group cohesion term was significantly associated with local learning ($\beta = -2.53$, $p < 0.05$), suggesting that the relationship between group cohesion and local learning may not be as simple as previously thought. The implications of these findings are discussed in the next section. Consistent with the findings of the omnibus null hypothesis test, task routineness was not significantly related to local learning ($\beta = 0.07$, $p > 0.10$) and distal learning ($\beta = -0.15$, $p > 0.10$). Hence, Hypotheses 4a and 4b were not supported.

Additional Analysis

Using Baron and Kenny's (1986) test of mediation, I examined whether the group learning processes mediate the relationships between group contextual conditions (i.e., group cohesion and task routineness) and group performance. Since task routineness was not significantly related to either distal or local learning, the mediation test was only performed for group cohesion. Controlling for the control variables, group cohesion was marginally significantly related to group efficiency ($\beta = 0.26$, $p = 0.51$) but not group innovativeness ($\beta = 0.13$, $p > 0.01$). In addition, as reported in Table 2, only local learning was significantly related to group efficiency ($\beta = 0.35$, $p < 0.05$). Therefore, only the mediating effect of local learning on the relationship between group cohesion and group efficiency was tested. When local learning was added to the regression model in which group cohesion predicted group efficiency, local learning was significantly related to group efficiency ($\beta = 0.31$, $p < 0.05$) and the coefficient for group cohesion became insignificant ($\beta = 0.12$, $p > 0.10$). Also, as reported in Table 3, group cohesion was significantly related to local learning ($\beta = 0.45$, $p < 0.001$). The results therefore indicated that local learning mediated the relationship between group cohesion and group efficiency.

Discussion and Implications

As the focus of a growing number of studies, both local learning (e.g., Fiol 1994, Orr 1990) and distal learning (e.g., Ancona and Caldwell 1992a, Hansen 1999) have been demonstrated to be central to group performance. However, the trade-offs and tensions associated with local and distal learning deserve greater theoretical and empirical attention than they have been given to

date. This study contributes to this area of research in the following three ways.

First, the findings reinforce our understanding that different types of group learning have different impacts on group performance. Distal learning was positively associated with group innovativeness, and local learning was positively associated with group efficiency, suggesting that there are performance trade-offs to selecting one type of group learning over the other. This pattern of findings parallels March's (1991) theory that while exploring distant knowledge bases improves innovativeness, exploiting familiar knowledge bases enhances efficiency. Notably, even though local learning involves engaging in exploratory-oriented behaviors such as trying new ideas and reflecting, the more circumscribed knowledge space likely to be accessed in local learning (i.e., only sampling the knowledge of individuals within the same group) serves to improve only the efficiency and not the innovativeness of the group. Given that distal and local learning are significant for different performance dimensions, the findings imply that learning-oriented teams with emphasis on developing new competencies (e.g., Bunderson and Sutcliffe 2003) should engage in greater distal learning, and mastery-oriented teams with emphasis on perfecting current competencies should engage in greater local learning.

Second, the findings of this study reframe the common belief that local and distal learning are mutually compatible learning processes by demonstrating the negative interaction effect of local and distal learning on efficiency. Studies on internal and external group communication (e.g., Katz and Tushman 1979) and intra- and intergroup knowledge exchange (e.g., Ancona and Caldwell 1992b, Hargadon 1999) have suggested that groups that engage in both internal and external knowledge sharing perform better. However, the findings of this study illustrate that distal learning interferes with local learning's ability to achieve efficient performance. Particularly, local learning becomes insignificantly related to group efficiency when distal learning is high. Therefore, for groups whose performance is primarily driven by the efficient execution of tasks, the findings of this study would suggest a focus on local learning alone and providing a buffer for the group from external influences.

Third, this study refines current understanding of the positive role of social relationships in promoting collective learning by showing that a high level of social cohesion may not always be beneficial for group learning. Specifically, the unexpected finding of an inverted U-shaped relationship between group cohesion and local learning suggests that excessive social integration in the group reduces mutual learning among members. However, in the case of distal learning, a higher level of group cohesion promotes greater external knowledge sharing. This counters the popular belief that higher

internal group cohesiveness fosters greater insularity and resistance to external ideas. These differential relationships of local and distal learning with group cohesion highlights one tension in fostering both types of group learning at the same time (e.g., Argote et al. 2001) by demonstrating that while a high amount of social cohesion promotes distal learning, it diminishes local learning. However, in light of the modest and associative nature of the findings (particularly, the relationship between distal learning and group cohesion), these findings should be interpreted with caution and it would be worthwhile for future research to further examine how group cohesion is related to different group learning processes.

Several insignificant findings in this study also bear further mention. Contrary to expectation, local learning did not complement distal learning to enhance group innovativeness. A possible reason is that range restriction—local learning scores of the teams in this sample were clustered toward the high end (the scores ranged from 2.7 to 4.2 on a scale from 1 to 5, with a mean of 3.45) with only about 13% of the teams with local learning scores below 3—reduced the likelihood of detecting a significant interaction effect. In addition, task routineness was not significantly related to either distal learning or local learning. The insignificant findings, combined with the significant associations of group cohesion with distal and local learning, may suggest that social conditions such as group cohesion, trust, and climate are more predictive of interpersonal learning processes (e.g., Edmondson 1999) than task conditions. However, given its speculative nature, it would be fruitful for future research to investigate whether other task conditions (e.g., task complexity) affect group learning.

Limitations and Future Research Directions

There are several limitations in this study that open avenues for future research. First, because this study focuses on interpersonal group learning processes, the differential relationships of local and distal learning with group performance and group cohesion are bounded by the interpersonal nature of the learning processes examined in this study. Knowledge, however, can also be acquired and shared in the absence of social interactions; for example, through accessing databases (Culnan 1983) and transferring tools developed in one group to another (Argote and Ingram 2000). Thus, future research could explore whether noninterpersonal means of distal learning (e.g., accessing external databases) would have the same group performance effects as those found in this study. By adopting a multifaceted perspective to learning (e.g., examining how embedded knowledge in members, tools, and tasks is shared), we might learn how different types of knowledge-sharing networks complement one another (Argote and Ingram 2000, Arrow et al. 2000).

Second, different knowledge effects were theorized to mediate the relationships of distal and local learning with group performance, but they were not empirically tested in this study. Given that there is considerable interest in the types of knowledge that are developed by different group learning processes, it would be worthwhile for future research to systematically examine the knowledge outcomes of distal and local learning.

Third, the cross-sectional nature of the dataset prohibits the testing of causality. Even though this research posits the influence of group cohesion on distal and local learning, it is likely that distal and local learning impact group cohesion. In the absence of longitudinal data, this research can only conclude associative, not causal, inferences about the relationships among the constructs. Hence, it would be valuable to conduct longitudinal research on how group contextual conditions foster group learning and how group learning might, in turn, shape the group context.

Finally, the self-reported nature of the data increases the risk of common method variance. In mitigation, using the Harman one-factor test (Podsakoff and Organ 1986), the risk of common method variance is unlikely to be significant in this case because the exploratory factor analytic results showed that one general factor did not account for most of the variance. Tests of discriminant analyses in the confirmatory factor analysis also evidenced that the constructs were distinct from one another. Group performance variables were obtained from the team manager, thereby decreasing the risk of percept-percept bias. While it is possible that managers' performance ratings were affected by their evaluation of the extent of group learning processes, thereby inflating the relationship between group learning and group performance variables, they are generally more likely to positively correlate the two when they experience these processes as impacting "objective" indicators of group performance. In a review of performance evaluations in work settings, Arvey and Murphy (1998) also found that there is "increased recognition that subjectivity does not automatically translate into rater error or bias and that ratings are most likely valid reflections of true performance" (p. 163). However, the risk of perceptual biases remains whenever subjective ratings are used, thus presenting a potential limitation of this study. It would be useful for future research to use objective measures of performance so as to provide a more robust test of the relationship between group learning and group performance.

Conclusion

Responding to Argote and associates' (2001) call for a more in-depth understanding of the trade-offs and tensions underlying different group learning processes, this

study presents a more fine-grained investigation of the relationships of local learning and distal learning with group performance, and group social and task conditions. The findings of this study not only provide empirical support for the notion that local and distal learning facilitate group performance in different ways, but also advance group learning theory by highlighting that the two types of learning processes can mutually interfere with one another such that group efficiency is worsened, and they can be difficult to foster simultaneously because a high level of social cohesion enhances distal learning but diminishes local learning.

From a practical standpoint, this study highlights to managers the importance of recognizing the trade-offs associated with distal and local learning and, hence, the value and applicability of each type of group learning. Depending on whether group innovativeness or group efficiency is the primary concern, distal and local learning vary in importance. Given the interaction effects of distal and local learning, managers should also be mindful of the delicate balancing act between distal and local learning. Finally, the findings of this study convey to managers the different implications of high social cohesion on the two types of group learning and, thus, highlight that different design and process interventions are required, depending on which group learning process is targeted.

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Appendix

Multi-item measures

Local learning

1. People in this team speak up to test assumptions about issues under discussion.
2. We create hypothetical (what-if) scenarios to achieve new insights.
3. We take time to figure out ways to improve our team's work processes.
4. Our team explores one another's ideas to invent new or better ways we might perform our task.
5. When a nonroutine matter comes up in our work, we consult one another's opinions about how to handle the situation.
6. When errors occur, we discuss how to prevent these errors in the future.
7. We reflect on our past ways of doing things.
8. When we encounter new issues, we discuss what these issues mean in our work.

Distal learning

1. Our team seeks ideas/expertise from people external to the team.
2. Our team reviews the team's work with people external to the team.
3. Our team obtains help or advice from people external to the team.
4. Our team seeks feedback about the team's work from people external to the team.

Group cohesion

1. We are ready to defend each other from criticism from outsiders.
2. We help each other on the job.
3. We get along well with each other.
4. The members in this team really stick together.

Task routineness

1. The situations, problems, or issues we encountered were similar from day-to-day.
2. This task resembles other tasks that we have to do.
3. We routinely see tasks of this type throughout our organization.
4. Most of the tasks we perform are similar from day-to-day.

Endnotes

¹In the pilot study, an eight-item local learning measure was used (the other three items were added after the pilot study). Based on a sample of 53 team members, results of the exploratory factor analysis showed 4 factors with eigenvalues greater than 1. All the factor loadings were greater than 0.40. Items from the frequency of communication scale and positive group interaction scale loaded on their respective factors. The local learning items loaded on another two factors, with three items loading on one factor (items reflecting more knowledge creating) and five items loading on another factor (items reflecting more knowledge integration). As shown later in the main study, the local learning items loaded on one factor when a larger sample was used.

²Notably, the overall F -ratios of the two regression models were not significant. As Cohen and Cohen (1983) highlighted, variables that account for little of the variance in the dependent variable, which in this case were the control variables, are likely to reduce the power of the statistical tests on the entire set of independent variables. Therefore, they recommended the use of Model I error (i.e., the F test for increment in Y variance from adding a new set of variables) at each level of the hierarchical regression analysis to be used as the test of significance for the set of variables entered at each level. Because the change in R^2 was significant when the distal and local learning variables were entered in the regression models, the interpretations of the individual significance of distal and local learning on group innovativeness and group efficiency are meaningful. In addition, when distal and local learning were regressed on group innovativeness and group efficiency (i.e., without the control variables), the F -ratios for group innovativeness ($R^2 = 8.6\%$, $p < 0.05$) and group efficiency ($R^2 = 10\%$, $p < 0.05$) were significant, further demonstrating that the control variables were lowering the statistical power of the F -tests of significance for both regression models.

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